LUIS PRATAMA

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PROFILE

Graduate Mechatronics Engineer with experience in mechanical systems design, prototyping, and system integration. Completed an industry internship and multiple engineering projects involving SolidWorks-based mechanical design, automation, and embedded systems. Experienced in technical documentation, testing, and multidisciplinary team collaboration.

ENGINEERING EXPERIENCE

Engineering Intern *May 2024 - August 2024*

Optik Consultancy / Byron Bay Technology

- Designed and created a functional nozzle housing for Byron Bay Technology, enabling multidirectional movement and auto shut off
- Modelled mechanism in SolidWorks with attention to tolerances, alignment, and precise fit
- Integrated servo motors and Arduino to automate nozzle control and shut-off operation
- Collaborated with engineers on prototyping, testing, and mechanical-electrical system integration
- Gained hands-on experience in embedded systems, motion control, and design for manufacturing
- Tools & Skills: SolidWorks, Arduino, Servo Motors, Mechanical Design, Embedded Systems, DFM, Prototyping

CASUAL EXPERIENCE

Hospitality Staff *May 2022 - Present*

Super Agent Agency

- Provide professional front-of-house support at high-profile hospitality events, ensuring smooth guest experiences and efficient service delivery
- Work collaboratively in dynamic team environments to coordinate event setup, food and beverage service, and breakdown
- Maintain high standards of presentation, communication, and time management across diverse event settings
- Adapt quickly to fast-paced and high-pressure situations, supporting events leads and client needs flexibility and reliability
- Developed strong interpersonal, multitasking, and customer service skills applicable across client-facing and team-based roles
- Skills: Communication, Teamwork, Time Management, Client Service, Adaptability, Event Coordination

Barista *May 2023 - Present*

MilkMe Barista

- Prepare and serve high-quality coffee and espresso beverages in a fast-paced mobile café environment across various Sydney events
- Operate commercial espresso machines with consistency and attention to detail.
- Maintain cleanliness, stock levels, and setup of mobile coffee stations in line with food safety and hygiene standards
- Deliver friendly and efficient service while managing high customer volume during peak hours
- Skills: Customer Service, Espresso Preparation, Time Management, Food Safety, Teamwork

RELATED PROJECT

UR3e Bottle Sorting Cobot - Perception System *February 2025 - May 2025*

Robotics Studio 2

- Built a ROS 2 node to capture and process RealSense camera images for color-based bottle-cap detection

- Created an OpenCV GUI with live sliders and click-to-sample tools for on-the-fly threshold tuning
- Designed a simple and adjustable interface to test color detection quickly and accurately
- Translated visual data into a robot-readable format
- Ensured system worked smoothly under different lighting and ran in real time
- Worked closely with team members responsible for robot motion and gripper to ensure everything worked together

SortBot Simulation - YOLO *February 2025 - May 2025*
Artificial Intelligence in Robotics
- Built a PyBullet simulation of a uArm robot sorting colored boxes using AI-powered vision and motion planning
- Collected and labeled 1,000+ top-down images to train a real-time object detector, achieving over 95% box-identification accuracy in simulation
- Evaluated robot detection and pick-and-place performance under varied conditions
- Improved how the system responded by mapping camera and robot positions accurately
- Presented progress regularly to peers and incorporated feedback from supervisors

Camera Guidance Mechanism for Small-Diameter Conduits (NBN) *August 2025 - November 2025*
Design in Mechanical and Mechatronics Systems
- Integrated a compact camera inspection system within a motorised conduit-traversal mechanism (Ø20–100 mm)
- Implemented Raspberry Pi–Arduino serial control for bidirectional motion using dual NEMA 17 stepper motors
- Developed a lightweight operator interface with live camera feed, motion commands, and image capture
- Designed and validated a protected 12 V power distribution system supporting ≥2 h continuous operation
- Supported prototyping, integration, and laboratory testing in simulated conduit environments

TECHNICAL SKILLS

Language	C/C++, Java, Phyton, Arduino
Systems	macOS, Linux (Ubuntu), Windows
Software	ROS 2, OpenCV, SolidWorks, AutoCAD, Fusion 360, MATLAB, Visual Studio Code

CORE SKILLS

Teamwork
- Collaborated with a multidisciplinary team of five engineers to develop and deliver a client-focused project.

Problem Solving
- Contributed to solution brainstorming for a sorting system during internship, with the proposed concept later developed into a prototype showcased in a client demo.

Communication
- Demonstrated strong communication skills by coordinating with both the client and team members, contributing to smooth workflow and successful project delivery.

Multitasking
- Managed multiple academic deadlines while maintaining part-time employment, demonstrating strong time management and personal responsibility.

Language
- English (IELTS - 7)
- Bahasa Indonesia (Native)

EDUCATION

Bachelor of Engineering in Mechatronics (Honours) **February 2022 - December 2025**
University of Technology Sydney (UTS)

Diploma of Engineering **March 2021 - December 2021**
University of Technology Sydney College

CERTIFICATION

- Responsible Service of Alcohol (RSA)
- Class C Driver's License

REFEREES

Available on request.